Daniel Martín Guerrero

BSc in Physics (2025–2029) · second-year undergraduate
Faculty of Sciences, Universidad de Salamanca
Optics & Photonics · Astrophysics · Quantum Mechanics · Science Communication

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Education

Universidad de Salamanca (USAL) – BSc in Physics *2025 – Present*
Second year, expected June 2029 · *Highest Honours in* Physics Laboratory Salamanca, Spain

- **Mathematics:** linear algebra and geometry (matrix and tensor calculus; bilinear, sesquilinear and quadratic forms; dual spaces; affine and Euclidean geometry; foundations of differential geometry); multivariable and vector calculus, series and complex variable; differential equations for physical models (logistic, Gompertz, SIS and Lotka–Volterra dynamics; RLC circuits via Laplace transforms; Fourier signal analysis); introduction to topology, stochastic processes and measure theory.
- **Physics:** classical mechanics, oscillations and waves (fundamentals of optics), electromagnetism, thermodynamics and computational techniques; uncertainty quantification (propagation of variances, weighted means, χ^2 least-squares fits, Welch–Satterthwaite coverage intervals).

High School Diploma (Bachillerato) – Science and Technology track *2023 – 2025*
IES Federico García Bernalt · *graduated* **with Honours**, *GPA* **9.88/10** · *PAU* **13.378/14** Salamanca, Spain

Science Communication & Outreach

Radio USAL – *Eureka*, science outreach programme *2026 – Present*
Scriptwriter & Presenter; creator of the segment Cosmosofía Supernova Astronomy Association

- Astronomy content for the University of Salamanca's official radio station. *Cosmosofía* bridges astrophysics and philosophy of science: cosmic coordinates, stellar nucleosynthesis, Cecilia Payne's discovery and Sagan's *Pale Blue Dot.*

European Researchers' Night 2026 – Universidad de Salamanca *25 Sep. 2026*
Activity leader, "Houston, tenemos un telescopio" *(Houston, we have a telescope)* Salamanca, Spain

- Ran a three-hour public astronomy activity in the EU-funded Marie Skłodowska-Curie European Researchers' Night, part of the 2025–2026 programme of the USAL Scientific Culture and Innovation Unit (UCC+i); officially certified.

TikTok – @_.danielmg *(content creator)* *Ongoing*

- AI-assisted physics and mathematics videos, from astrophysics, electromagnetism and thermodynamics to Sylvester's law of inertia, linear differential forms and the generalised Stokes theorem.

Talks & Conferences

XXX International Congress of the Castilian-Leonese Society of Philosophy, *Límites del presente* *19–21 Nov. 2025*
Speaker: "Como lágrimas en la lluvia" *(Like Tears in Rain)* USAL, Salamanca

- Session *Anthropocentrism and androcentrism as a limit of compassion*, coordinated by Prof. Angélica Velasco Sesma (Univ. of Valladolid).

XXI Philosophical Olympiad of Castilla y León, *Madness: Don Quixote in Today's World* *27 Feb. 2026*
Invited Speaker & Round-Table Panellist USAL, Salamanca

- *"The Shipwreck of Light: on the lucidity of delirium – Don Quixote, Myshkin and the aesthetics of goodness"*.

Book presentation of *Salvar a Sócrates* (Ariel) with author Eduardo Infante *5 Jun. 2026*
Panellist Librería Víctor Jara, Salamanca

Projects & Independent Study

***Talking Hands* – Sign-Language Translator Glove** *2024 – 2025*
Hardware & firmware co-developer (with Rubén González Sánchez), ICT II project IES Federico García Bernalt

- Built a wearable assistive device that translates the Spanish fingerspelling alphabet into text, so that people with speech impairments can communicate with those who do not know sign language.
- Instrumented a glove with five flex sensors in voltage-divider circuits on an Arduino Mega 2560 (10-bit ADC); wrote C/C++ firmware for sensor calibration and threshold-based gesture classification, streaming letters over serial to a mobile app for display and speech.
- Added a servo mode mirroring finger flexion, a 3D model and a proposed wireless version with skin-like pressure sensors.
- **Quantum Mechanics & Computing:** authored LaTeX notes on Hilbert spaces and the Dirac formalism ($L^2(\mathbb{R})$, bras and kets, Hermitian operators, spectral decomposition, Born rule); studying qubits and quantum gates. *2026 – Present*
- **AI-Assisted Modelling:** physical simulations, from parametrised manifolds to least-squares fits, and study notes with Claude Opus 5.5, Gemini, open-source LLMs and NotebookLM. *2025 – Present*

Associations & Memberships

- **Supernova**, University Astronomy Association (USAL): outreach events, observation nights, retreats and radio. *2025 – Present*
- **OSAL Student Chapter**: student association for optics and photonics outreach, Universidad de Salamanca. *Member*
- **Optica** (Advancing Optics and Photonics Worldwide): Individual Member. *Through Sep. 2027*
- **Delegación Charra**: Salamanca delegation of the Student Group of the Royal Spanish Physics Society (GdeE–RSEF). *Member*

Honours & Awards

- **University Access Award**, USAL: top 100 scores in the general phase of the 2024/25 PAU; full first-year tuition waiver. *2025*
- **Finalist**, Philosophical Olympiad of Castilla y León, philosophical dissertation (XIX and XX editions). *2024, 2025*
- **Participant**, PLANCKS 2027 – international theoretical physics competition. *Upcoming*

Technical Skills & Interests

Programming: Python, MATLAB, Arduino (C/C++ on the Wiring framework), HTML (basic).
Lab & tools: Breadboard circuit prototyping with Arduino and LabVIEW; Mathematica, PHYWE measure, SAOImage DS9 (astronomical image analysis), LaTeX (advanced).
Languages: Spanish (native); English (advanced, C1-level training).
Interests: Astrophysics, optics and photonics, particle physics; philosophy and cinema (Tarkovsky, Lynch, Bergman, Kurosawa).